26900559 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

Regular EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEZGYA02 - ADVANCED ECOLOGICAL CONCEPTS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. R-selected organisms generally produce: [U] / [CO2]

   a). Few offspring b). Many offspring

   c). No offspring d). One offspring only

2. The basic unit of ecological study is the -----. [K] / [CO1]

   a). Population b). Community

   c). Organism d). Ecosystem

3. The division of a continuous habitat into smaller, isolated patches is known as [U] / [CO3]

   a). Habitat succession b). Habitat fragmentation

   c). Habitat restoration d). Habitat colonization

4. Annual plants are commonly associated with: [K] / [CO2]

a). Iteroparity b). Semelparity

c). Polyembryony d). Parthenogenesis

5. Ocean acidification primarily causes a decrease in seawater: [K] / [CO3]

a). Temperature b). Salinity

c). pH d). Oxygen concentration

6. A key output of PVA is [K] / [CO3]

a). Species count b). Extinction probability

c). Taxonomic classification d). Evolutionary history

7. Which of the following is a major group of plant chemical defences? [U] / [CO2]

a). Proteins b). Alkaloids

c). Carbohydrates d). Lipids

8. Deterministic models are most suitable for studying [U] / [CO3]

a). Populations with unpredictable environmental fluctuations b). Large populations where random events average out

c). Very small populations facing extinction risk d). Genetic drift in isolated populations

9. Which of the following doesnot contribute to Habitat Fragmentation? [K] / [CO2]

a). Movement of organisms b). Deforestation

c). Urbanization d). Road construction

10. What is a metapopulation? [U] / [CO2]

a). A population of different species b). A population of populations

c). A single large population d). A community of organisms

[1x10 = 10]

Part B

Short Answer

Answer any **14** questions

Each question carries **2** marks

11. What is the n-dimensional niche concept? [U] / [CO1]

12. Define aggressive mimicry with one example [U] / [CO2]

13. What is phenological change? Give one example. [K] / [CO3]

14. Explain why stratified sampling is suitable for heterogeneous habitats and state any two advantages of this sampling method. [U] / [CO4]

15. What is the uses of Demographic techniques in studying Population Ecology ? [K] / [CO2]

16. Explain the major threats to biodiversity hotspots. [U] / [CO3]

17. What does relative abundance indicate? [K] / [CO2]

18. Explain how human activities contribute to the increase in greenhouse gas concentrations. [A] / [CO3]

19. Explain the importance of advanced ecology in climate change studies. [U] / [CO1]

20. Describe the Keystone and Idiosyncratic hypotheses of species richness and community services [U] / [CO2]

21. How can Exponential Growth Model of population represented mathematically? [K] / [CO2]

22. Why did vulture populations decline drastically in India? [U] / [CO4]

23. What is landscape connectivity? [An] / [CO3]

24. Differentiate between ectoparasites and endoparasites [U] / [CO2]

25. What is a keystone species? Give one example. [An] / [CO3]

26. What are the characteristics of the Logistic Growth Model of population? [K] / [CO2]

[2x14 = 28]

Part C

Short Essay

Answer any **8** questions

Each question carries **4** marks

27. Explain pollination syndromes and discuss how floral traits are adapted to attract specific pollinators [U] / [CO2]

28. What is the Lotka-Volterra model used for in ecology? [An] / [CO3]

29. Discuss the importance and limitations of adaptation and mitigation strategies for biodiversity conservation. [E] / [CO3]

30. Explain how Tokeshi's niche models help in understanding species abundance patterns in ecological communities [An] / [CO2]

31. What is remote sensing? Explain its integration with GIS in landscape monitoring. [An] / [CO3]

32. Describe the role of density-dependent factors in population regulation. [K] / [CO2]

33. Compare camera traps, GPS tracking, environmental DNA (eDNA), and bioacoustic monitoring as modern tools for ecological research and wildlife conservation. [An] / [CO4]

34. Calculate the Simpson Index for the above grassland community. Interpret whether the ecosystem shows high or low diversity, and explain which species contributes most to dominance. Antelope = 120, Zebras = 80, Lions = 20, Hyenas = 30, Vultures = 50 [An] / [CO2]

35. Describe resource partitioning and explain how it promotes coexistence among competing species [U] / [CO2]

36. Describe the role of habitat corridors in maintaining landscape connectivity. [An] / [CO3]

[4x8 = 32]